

22AIE111 OBJECT ORIENTED PROGRAMMING IN JAVA

S2 B.Tech CSEAI

Department of Computer Science and Engineering

Amrita School of Computing, Amritapuri Campus

CASE STUDY

PHASE – 2 (Design)

Architectural Modelling (CO2)

Group No : 21

Team Members :

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1. Problem Title : **Water Supply Management System**

STRUCTURAL DESIGN

2. Class diagram (draw class diagram that shows different classes and their relationships. Use UML notations)



BEHAVIORAL DESIGN

3. Use-case diagram

a. List of actors

1. Consumer
2. Operator
3. Admin
4. System

b. List of use-cases

1. **Consumer:**

- Use Water – Consumer consumes water.
- View Usage – Consumer checks usage.
- Pay Bill – Consumer pays bill.
- Report Leakage – Reports issues.

2. **Operator:**

- Control Water Supply – Manages distribution.
- Update Schedule – Updates timings.
- Monitor Flow – Observes flow.
- Stop Supply – Stops supply.

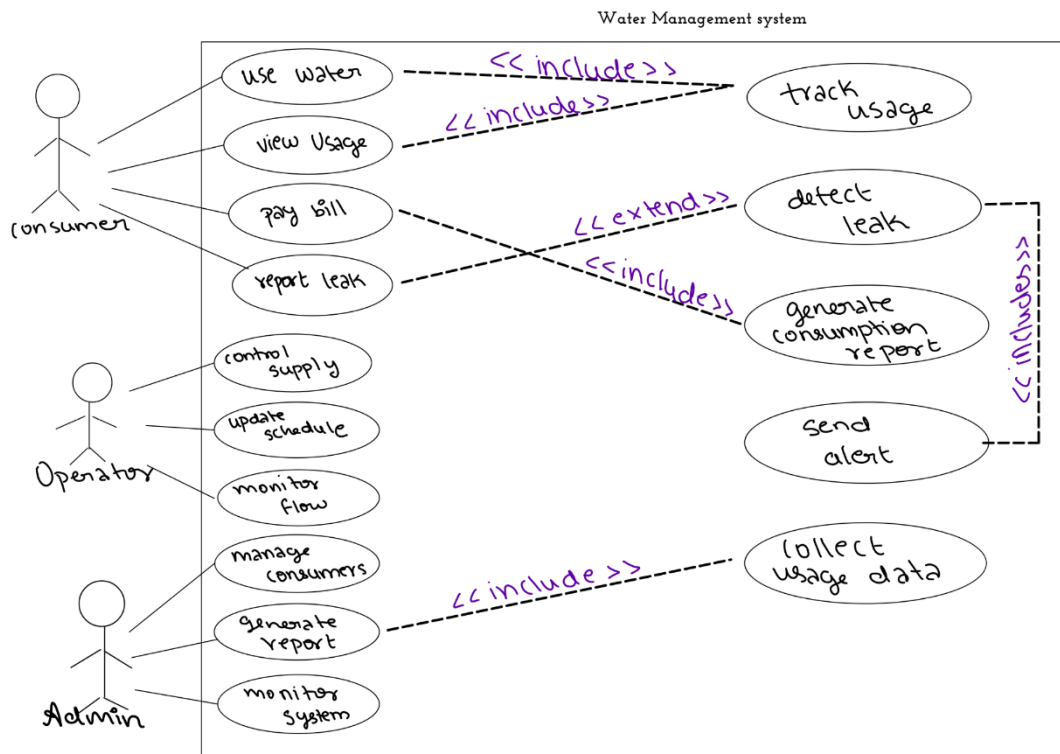
3. **Admin:**

- Add Consumer – Registers users.
- Remove Consumer – Deletes users.
- Generate Reports – Creates reports.
- Monitor System – Oversees system.

4. **System:**

- Track Usage – Records usage.
- Detect Leakage – Finds leakage.
- Generate Report – Creates consumption reports.
- Send Alerts – Sends notifications.

c. Use case diagram (use UML notations)



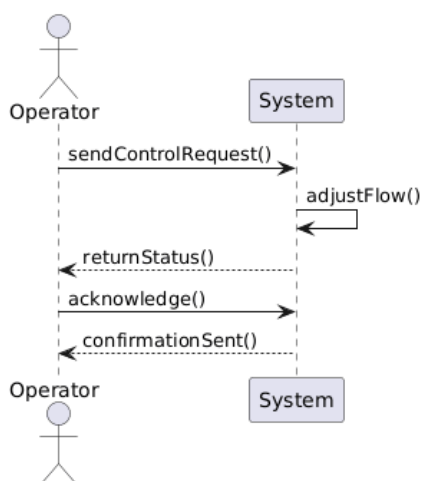
4. Sequence diagram (use UML notations)

- a. Choose key use cases and draw their sequence diagram. First describe the scenario in few sentences. Then draw the sequence diagram.

1. Use Case: Control Water Supply

Steps:

1. Operator sends control request
2. System adjusts flow
3. Status returned
4. Confirmation sent.

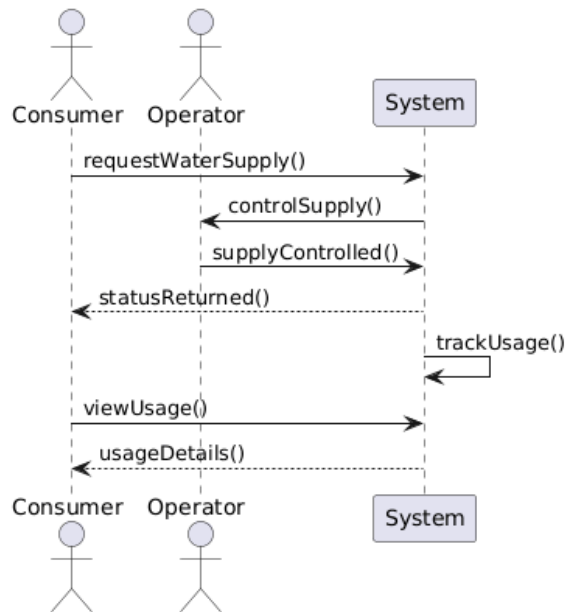


2. Use Case: Use water supply

Steps:

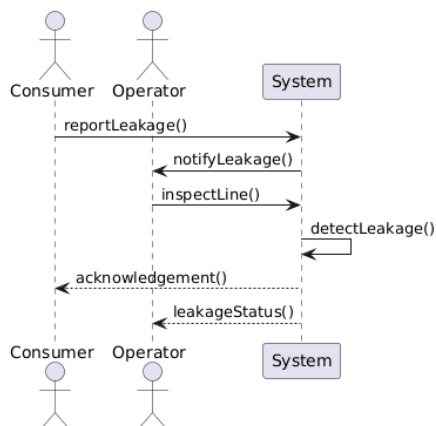
1. Consumer sends request.

2. Operator controls flow.
3. Status returned.
4. System tracks usage.
5. Customer views usage.



3. Report Leakage

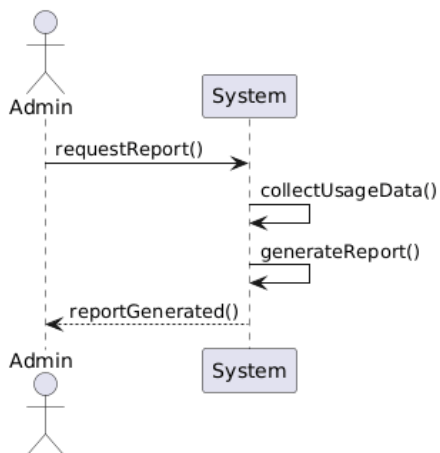
1. Consumer detects leakage.
2. System records the complaint.
3. Operator is notified.
4. Operator checks the supply line.
5. System updates leakage status.



4. Generate Report

1. Admin requests a report.

2. System collects usage and billing data.
3. System compiles the report.
4. Admin views the report.



5. Pay Bill

1. System generates the bill based on usage.
2. Consumer views bill details.
3. Consumer makes payment.
4. System confirms payment.
5. Payment status is updated

